

Applied Physics-I 1003

Semester 1 | Credits: 3 | Author: Nandakumar.M

This printable PDF is a compact revision copy. The full bilingual notebook page is [lessons/lessons-1003.html](https://www.nandakumar.com/lessons/lessons-1003.html).

Study method: read concept, write formula, symbol meaning, SI unit, example and application.

## Module 1 - Measurements, Vectors and Motion

Physical quantity: measurable property. Fundamental quantities include length, mass and time. Derived quantities include velocity, force and pressure.

Errors: absolute error =  $|\text{measured} - \text{true}|$ ; relative error = absolute error / true value; percentage error = relative error  $\times 100$ .

Vectors:  $A_x = A \cos \theta$ ,  $A_y = A \sin \theta$ ,  $R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$ .

Motion:  $v = u + at$ ;  $s = ut + \frac{1}{2}at^2$ ;  $v^2 = u^2 + 2as$ .

Newton laws: inertia,  $F = ma$ , action and reaction. Momentum  $p = mv$ . Impulse =  $Ft$ .

Momentum conservation explains recoil of gun and rocket propulsion.

## Module 2 - Circular Motion and Rotation

Angular quantities:  $\omega = \theta/t$ ,  $\alpha = \text{change in } \omega/\text{time}$ . Relation:  $v = r \omega$  and  $a = r \alpha$ .

Centripetal acceleration  $a_c = v^2/r = r \omega^2$ . Centripetal force  $F_c = mv^2/r$ .

Banking of roads:  $\tan \theta = v^2/rg$ . Banking reduces skidding on curves.

Moment of inertia:  $I = \sum mr^2$ . Radius of gyration:  $I = Mk^2$ .

Common MI: rod centre  $ML^2/12$ , rod end  $ML^2/3$ , ring  $MR^2$ , disc  $MR^2/2$ , solid sphere  $2MR^2/5$ .

Torque =  $rF \sin \theta$ . Angular momentum  $L = I \omega$ . If external torque is zero, angular momentum is conserved.

### Module 3 - Work, Energy, Power and Heat

Work  $W = Fs \cos \theta$ . Work is zero if displacement is zero.

Friction  $f = \mu R$ . Reduce friction using lubrication, polishing, ball bearings and streamlining.

Kinetic energy  $KE = \frac{1}{2} mv^2$ . Potential energy  $PE = mgh$ . Energy is transformed, not destroyed.

Power  $P = W/t = Fv$ . Unit is watt.

Temperature conversion:  $K = C + 273$ ;  $F = \frac{9C}{5} + 32$ . Heat  $Q = mc \Delta T$ .

Heat transfer: conduction in solids, convection in fluids and radiation without medium.

Mercury thermometer uses expansion; pyrometer measures high temperature without contact.

#### Module 4 - Elasticity, Pressure and Fluids

Elasticity: body regains original shape after removing deforming force. Stress =  $F/A$ .

Strain = change/original. Young modulus  $Y$  = stress/strain.

Pressure  $P = F/A$ . Absolute pressure = atmospheric pressure + gauge pressure.

Surface tension  $T$  = force/length. Capillary rise  $h = 2T \cos \theta / (\rho g r)$ . Detergents reduce surface tension.

Viscosity is resistance between fluid layers. Stokes law:  $F = 6 \pi \eta r v$ .

Reynolds number  $Re = \rho v D / \eta$ . Low  $Re$ : streamline flow; high  $Re$ : turbulent flow.

Continuity equation:  $A_1 v_1 = A_2 v_2$ . Bernoulli:  $P + \frac{1}{2} \rho v^2 + \rho g h = \text{constant}$ .

Applications: airfoil, atomiser, venturimeter.

### Solved Problems and Practice

1. Rod measured 2.04 m instead of 2.00 m: absolute error = 0.04 m, relative error = 0.02, percentage error = 2%.
2. Body from rest,  $a = 2 \text{ m/s}^2$ ,  $t = 5 \text{ s}$ :  $v = 10 \text{ m/s}$ ,  $s = 25 \text{ m}$ .
3. Wheel radius 0.5 m,  $\omega = 20 \text{ rad/s}$ :  $v = r \omega = 10 \text{ m/s}$ .
4. Force 20 N moves body 5 m: work = 100 J.
5. Stress for 100 N on  $0.02 \text{ m}^2$ : stress = 5000 Pa.

Expected questions: Newton laws, vector resolution, banking, MI, conservation of energy, heat transfer, Hooke law, surface tension, Bernoulli theorem.

Footer

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Expected questions are practice-focused, not a guarantee of actual exam questions.